



SENTIENT TOKEN AND ECONOMIC OPTIMIZATION FOR AGENTS

Make agent optimization falsifiable.

An open operation-control and evidence layer that binds what an agent was asked to do, what actually ran, what it cost, and whether an external verifier accepted the result.

PUBLIC-GOODS REQUEST

\$150,000

Nine months. Three go/no-go tranches. No equity. Open methods, code, evidence, and negative results.

809

public GitHub stars

24

outside-authored repositories

32 / 32

official holdout verdicts

OPEN

MIT code and public proof

PROBLEM

The model call is not the economic unit.

Real cost and failure accumulate across the entire operation.

- Decomposition, parallelism, retries, tools, context, recovery, and verification can overwhelm a cheap model choice.
- A requested model can differ from the provider path that actually executes.
- Unknown local, subscription, setup, or human cost is often collapsed into zero.
- Model prose or stack status can be promoted to completion without an external artifact check.

01 Declared plan

Models, tools, topology, and limits

02 Observed execution

Calls, attempts, artifacts, and cost lenses

03 External verdict

Deterministic verifier outside the route

04 Signed evidence

Offline-reconstructable receipt chain

UNKNOWN STAYS UNKNOWN



WORKING SUBSTRATE

Built before the grant.

Citadel starts with one command, then exposes stronger control only when the operation needs it.

- /do

Beginner path

Natural-language routing, verification, and a durable next action.
- PLAN

Portable contract

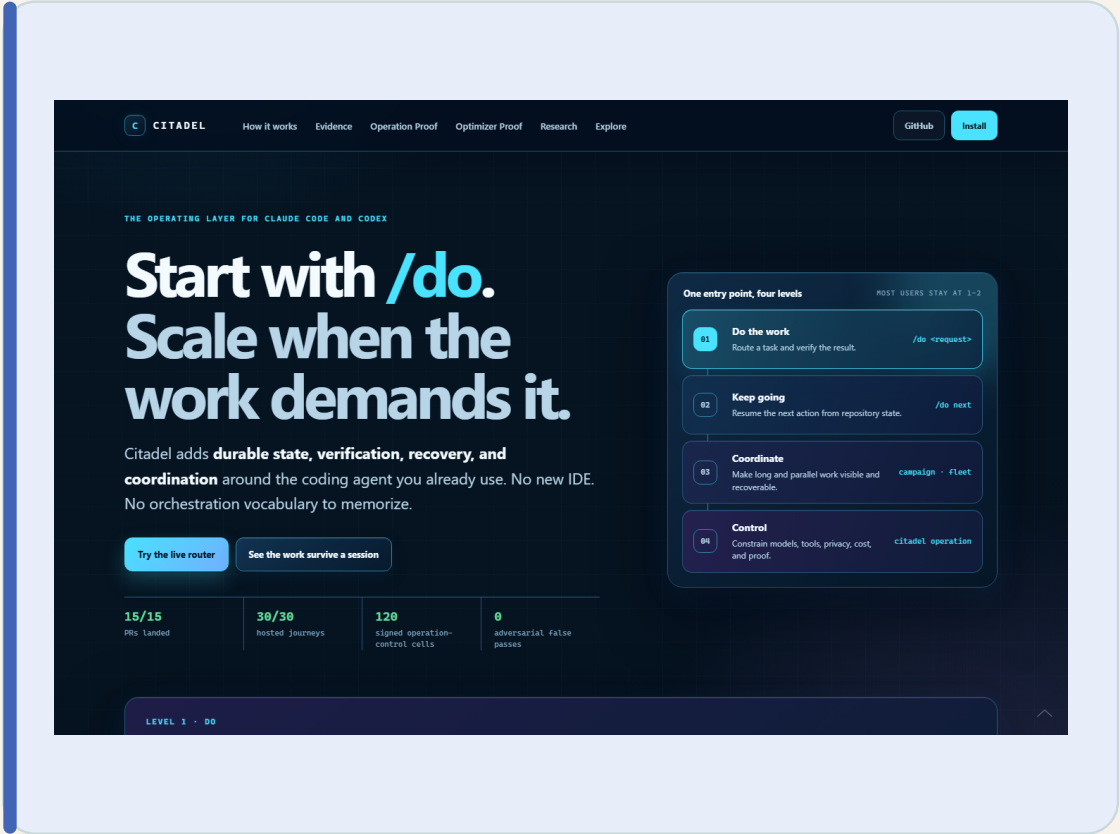
Models, tools, topology, retries, privacy, time, cost, and stop rules.
- RUN

Runtime coverage

Claude Code, Codex, Ollama, and a pinned Sentient ROMA binding.
- PROOF

Evidence plane

External artifact verdicts, failure preservation, signed receipts, and offline replay.



OUTSIDE-AUTHORED REPOSITORY DIAGNOSTIC

The evidence worked. The baseline did not.

Twenty-four distinct repositories. Routes sealed before evaluation. Every official verdict published, including the result that weakens the proposal.

3 / 16

sealed controller verified

2 / 16

direct Claude verified

1 / 16

local Qwen verified

-1.26%

comparison cost; not cash

WHAT COMPLETED

- Eight calibration and sixteen untouched evaluation tasks across four strata.
- Sixteen signed routes merged before model calls; 32/32 official verdicts.
- No task replacement, optional stopping, or positive-only publication.

WHY THIS IS NOT A WIN

Direct Claude passed only 12.5% overall and 0% in three strata. Matching or slightly exceeding that baseline does not establish useful quality preservation or general savings.

BASELINE VALIDITY FAILED



BUILT BEFORE FUNDING

One maintainer already shipped the substrate.

Citadel is a public system with continuing repository pull, not a grant-funded mockup.

136

days public

514 / 538

main commits attributed to Seth

809 / 80

stars / forks

524

unique cloners in 14 days

WORKING SUBSTRATE

- Claude Code, Codex, Ollama, and a pinned Sentient ROMA binding.
- 49 workflows and 35 hooks across 29 lifecycle events.
- Windows, Linux, and macOS verification on every release path.

CLAIM DISCIPLINE

A synthetic support envelope passed 12/12 at 38.7% lower comparison cost. The outside-authored follow-up then invalidated its strong-baseline claim. Both remain public.

FUNDING SCALES A WORKING BASE

NINE-MONTH RESEARCH AND ENGINEERING PROGRAM

Funding buys a valid operation baseline.

Repair the retrieval and edit substrate exposed by outside-authored work, then rerun the economic question.

01	Valid workload and baseline	Retrieval/edit contract, 60+ operations, complete cost boundary	\$30k
02	Adapter SDK	Sentient ROMA plus two additional open stacks	\$28k
03	Expected-value controller	Price verification, escalation, recovery, and failure	\$35k
04	Prospective evaluation	Frozen multi-stack policy and statistical contract	\$32k
05	Public proof and release	Hosted reconstruction, /do onboarding, support window, and maintenance	\$25k



NO-GO RULES ARE PART OF THE PRODUCT

Capped spend. Hard outcome gates.

COST BASIS

Maintainer labor	\$99,000
Evaluation compute and tools	\$27,000
Hosted verification and retention	\$7,000
Reference hardware and measurement	\$7,000
Compatibility and accessibility costs	\$2,000
Bounded contingency	\$8,000

Total **\$150,000**

PRIMARY PROSPECTIVE GATE

> = 80%

absolute verified completion

> = 95%

of valid frontier rate

> = 30%

lower measured end-to-end cost

ZERO

false passes and integrity failures

UNKNOWN COST CANNOT COUNT AS SAVINGS

DIRECT FIT WITH A NAMED REQUEST FOR PRODUCTS

An open optimizer that can prove when it is wrong.

WHY FUND CITADEL

- Directly targets Sentient's Token and Economic Optimization request.
- Already binds a pinned Sentient ROMA stack without replacing its planner.
- A bounded synthetic pass plus a failed outside baseline identify exactly what funding must repair.

WHAT STAYS PUBLIC

- MIT code, adapters, contracts, methods, verifiers, cells, and reports.
- No mandatory Citadel account, token, hosted evidence endpoint, or telemetry resale.
- Users keep project state, operations, receipts, and the right to run locally.

Seth Gammon - solo builder and maintainer

Evaluator: github.com/SethGammon/Citadel/blob/main/docs/grants/EVALUATOR_START_HERE.md

OPEN. VERIFIABLE. FALSIFIABLE.